Does Shrinking a Model Make Its Predictions Less Trustworthy for Some Groups?

Group-Wise Calibration in Pruned and Quantized Tabular Models

Abstract

Machine learning models are often compressed through pruning and quantization so they can run on small, low-cost devices. Compression is usually judged by accuracy and overall confidence quality, but these averages can hide problems for smaller groups of people. This study asks whether a compressed tabular model's predicted probabilities become less reliable for some groups than for others. We compress Random Forests, gradient boosting models, and small neural networks at several levels. We then measure calibration (how well predicted probabilities match real outcomes) overall and within each demographic group, using four public datasets such as Adult, German Credit, Bank Marketing, and Credit Default. We also test whether simple recalibration after compression restores reliability. Results are averaged over multiple random seeds and checked with statistical tests. The goal is practical guidance on when compressed models are safe to deploy and when group-level checks are needed.

Datasets

Dataset	Size	Why it fits
Adult (Census Income)	48,842 instances, predicting whether income exceeds $50K	Large and standard, so groups are big enough for stable group-wise results
Statlog German Credit	1,000 instances, 20 features, good or bad credit risk	Small, so it shows what compression does when data is scarce
Default of Credit Card Clients	about 30,000 rows, predicting default next month	Mid-sized, a different domain, and has several group attributes uciuci